

PIERRE, SD, USA

WMO: 726560

Lat: 44.381N

Lon: 100.286W

Elev: 531

StdP: 95.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24025

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.3	-21.6	-27.7	0.3	-23.8	-25.2	0.4	-21.2	15.6	-7.1	14.4	-6.0	3.6	310	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	36.0	21.6	33.7	21.3	31.6	20.7	24.2	31.0	23.1	29.9	22.1	28.9	6.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.2	27.6	21.1	16.9	26.8	20.0	15.7	25.7	76.2	30.9	71.7	29.8	67.6	29.1	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	11.9	10.6	-28.5	40.0	3.6	2.4	-31.1	41.8	-33.2	43.2	-35.3	44.6	-37.9	46.4		
(5)			-28.7	26.2	3.6	1.2	-31.2	27.1	-33.3	27.8	-35.4	28.5	-38.0	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-7.6	-5.6	1.0	7.9	14.1	20.1	24.0	22.8	17.8	9.4	1.4	-5.4		
	DBStd	12.46	7.68	7.79	7.53	5.73	4.72	3.87	3.64	3.53	4.97	5.36	6.13	6.73		
	HDD10.0	2210	544	436	288	104	16	0	0	0	3	76	264	479		
	HDD18.3	4100	803	669	538	316	144	26	3	6	68	280	509	737		
	CDD10.0	1629	0	0	9	41	145	302	436	396	238	58	4	0		
	CDD18.3	475	0	0	0	2	15	77	181	143	53	4	0	0		
	CDH23.3	5176	0	0	8	50	211	792	1956	1465	629	64	1	0		
	CDH26.7	2294	0	0	1	13	60	291	965	678	266	21	0	0		
(14) Wind	WSAvg	4.8	4.8	4.8	5.2	5.5	5.2	4.6	4.3	4.3	4.7	5.0	4.8	4.8		
(15) Precipitation	PrecAvg	481	12	13	26	51	76	90	66	44	43	34	15	12		
	PrecMax	809	58	59	95	162	184	185	163	132	176	137	66	30		
	PrecMin	181	0	0	1	5	14	12	5	1	0	0	0	0		
	PrecStd	123	10	12	20	37	40	47	43	31	38	30	15	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	16.0	23.9	28.0	31.4	35.6	40.8	38.0	36.8	29.5	20.8	12.4		
		MCWB	5.8	7.7	11.9	14.9	19.4	22.3	21.9	21.2	20.1	16.5	11.2	5.7		
	2%	DB	7.2	11.0	18.8	23.9	28.1	32.1	36.7	35.1	32.6	24.3	16.3	8.0		
		MCWB	3.2	5.2	10.0	13.4	17.7	21.4	21.9	21.6	19.4	14.5	8.7	3.7		
	5%	DB	4.6	7.3	15.0	20.6	25.7	29.9	34.1	32.8	29.6	21.0	12.9	5.2		
		MCWB	1.8	3.7	8.3	11.6	16.6	20.3	21.8	21.1	18.5	12.8	7.1	2.0		
	10%	DB	2.7	4.4	11.6	17.5	22.9	27.9	31.8	30.5	27.1	17.8	9.9	3.1		
		MCWB	0.6	1.8	6.8	9.7	15.2	19.5	21.4	20.3	17.6	11.4	5.5	0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	8.6	13.0	16.2	21.2	25.0	25.8	25.1	22.5	18.1	11.7	6.5		
		MCDB	11.7	14.6	20.3	25.4	28.4	31.3	32.5	32.0	29.5	25.7	19.1	11.4		
	2%	WB	3.6	5.7	10.8	13.9	19.3	23.2	24.6	23.7	21.2	15.7	9.4	3.9		
		MCDB	6.9	10.2	17.6	22.7	26.2	29.7	31.5	30.3	29.1	22.4	15.7	7.7		
	5%	WB	1.9	3.7	8.8	12.1	17.5	21.8	23.5	22.6	20.1	13.7	7.4	2.2		
		MCDB	4.3	7.1	14.8	19.1	23.8	28.0	30.5	29.1	27.4	19.8	12.5	5.0		
	10%	WB	0.6	1.9	6.6	10.6	15.9	20.5	22.5	21.5	18.8	11.9	5.6	0.8		
		MCDB	2.6	4.3	11.5	16.6	21.7	26.3	29.6	28.2	25.4	17.1	9.6	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	12.0	12.4	13.3	13.0	13.0	14.3	14.1	14.3	12.5	11.6	10.3		
		MCDBR	13.4	15.4	18.6	18.8	16.8	15.7	17.0	17.1	18.2	17.8	16.3	13.5		
		MCWBR	9.9	10.4	10.6	9.8	8.8	7.3	6.2	6.8	7.5	9.2	9.9	9.6		
		MCDBR	13.1	15.0	17.9	17.5	15.0	13.8	14.8	14.8	15.6	15.7	15.8	13.0		
	5% WB	MCWBR	10.1	10.3	10.6	9.6	8.6	7.5	6.8	7.2	7.3	8.9	9.9	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.270	0.293	0.340	0.362	0.373	0.389	0.386	0.349	0.306	0.273	0.259		
	taud		2.514	2.510	2.519	2.380	2.394	2.394	2.378	2.368	2.450	2.541	2.542	2.510		
	Ebn at Noon		886	938	953	923	908	896	875	866	876	879	862	852		
	Edh at Noon		66	80	91	114	116	116	117	114	97	77	64	60		
(44) All-Sky Solar Radiation	RadAvg		1.88	2.84	3.89	4.93	5.94	6.72	6.89	5.76	4.59	3.02	2.02	1.52		
	RadStd		0.16	0.24	0.30	0.37	0.56	0.43	0.26	0.45	0.30	0.37	0.21	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page